# Sales Performance Analysis (SQL • PostgreSQL • Power BI)

## EXECUTIVE SUMMARY

This project analyzes 9,994 retail transactions to evaluate business performance across revenue generation, profitability, product categories, regions, and seasonal demand trends.

Using PostgreSQL, key performance metrics were calculated and structured into reusable analytical views.

The analysis reveals that the West region and Technology category drive the strongest financial performance, while excessive discounting and specific product lines significantly reduce profitability.

Seasonal demand spikes in Q4 indicate opportunities for strategic inventory and marketing planning.

These insights will be operationalized through an interactive Power BI dashboard to support data-driven business decision-making.

## PROJECT OVERVIEW

This project analyzes retail sales performance using PostgreSQL and Power BI. The goal is to identify key business patterns across revenue, profit, product categories, regions, and time trends, and to translate the findings into actionable business insights.

The analysis uses transaction-level retail sales data to explore overall sales performance, high-performing regions and categories, top-selling products, and profitability trends. The results will later be visualized through an interactive Power BI dashboard.

## DATASET

- Dataset: Superstore Sales Dataset

- Rows: 9,994
- Columns: 21

## OBJECTIVES

- **What is the total revenue generated by the business?**
- **What is the total profit generated by the business?**
- **Which regions generate the highest sales?**
- **Which product categories perform best in terms of sales and profit?**
- **What are the top-selling products?**
- **How do monthly sales trends change over time?**
- **Which products or categories show low or negative profitability?**

## TOOLS USED

- PostgreSQL
- pgAdmin
- SQL
- VS Code
- GitHub
- Notion
- Power BI

## PROCESS (SQL WORKFLOW)

1. The retail sales dataset was downloaded and inspected in Excel.
2. Extra non-transaction rows appended to the CSV were identified and removed.
3. The cleaned dataset was imported into PostgreSQL using pgAdmin.

4. SQL queries were used to calculate revenue, profit, regional performance, category performance, product-level performance, and time-based sales trends.
5. The analytical findings will be visualized using Power BI through an interactive business dashboard.

# KEY SQL QUERIES

## SQL Views Created

The following SQL views were created to make the analysis reusable and dashboard-ready:

- `v_sales_kpis` — stores total revenue and total profit
- `v_regional_performance` — summarizes sales and profit by region
- `v_category_performance` — summarizes sales and profit by category
- `v_subcategory_profitability` — identifies high- and low-performing sub-categories by profit
- `v_monthly_sales_trend` — summarizes monthly sales and profit trends over time
- `v_discount_analysis` — evaluates the impact of discount levels on sales and profitability

These views improve structure, enable modular analysis, and make it easier to connect PostgreSQL outputs directly to business intelligence dashboards such as Power BI.

# DASHBOARD

An interactive Power BI dashboard was developed to visualize revenue trends, profitability drivers, regional performance, and product-level insights.

The dashboard enables business stakeholders to explore performance dynamically using filters and drill-down capabilities.

## Dashboard Overview

An interactive Power BI dashboard was developed to translate SQL-based analytical findings into business-ready visual insights.

The dashboard presents a structured performance view beginning with executive KPIs (Total Revenue and Total Profit), followed by regional and category performance comparisons.

Users can dynamically explore the dataset using interactive filters for **Month, Region, and Category**, enabling deeper investigation into profitability drivers and seasonal demand patterns.

Key dashboard components include:

- Revenue and Profit KPI cards for executive monitoring
- Regional sales and profitability comparison charts
- Category-level performance visualization
- Profitability analysis by discount level
- Monthly revenue trend visualization (2015–2018)
- Sub-category profitability ranking
- Identification of loss-making products

The dashboard supports data-driven decision-making by allowing business stakeholders to quickly identify growth opportunities, pricing risks, and operational inefficiencies.

# KEY INSIGHTS

## Regional Sales Performance Analysis

Data Output Messages Notifications

| | region text | total_sales numeric | total_profit numeric |
|---|---|---|---|
| 1 | West | 725457.82 | 108418.45 |
| 2 | East | 678781.24 | 91522.78 |
| 3 | Central | 501239.89 | 39706.36 |
| 4 | South | 391721.91 | 46749.43 |

Sales performance varies significantly across regions.

The **West region generates the highest revenue at $725K and also delivers the strongest profitability at $108K**, making it the most important market for the business.

The **East region follows with $679K in revenue and $91K in profit**, indicating strong but slightly lower performance compared to the West.

The **Central region contributes $501K in sales but produces relatively low profit ($39K)**, suggesting potential pricing pressure or higher operational costs.

The **South region records the lowest revenue at $392K with $46K profit**, highlighting a potential growth opportunity for market expansion or targeted sales strategy.

These findings suggest the company should:

- prioritize investment and inventory allocation in the West and East regions
- investigate cost drivers affecting profitability in Central
- explore demand stimulation strategies in the South

## Category Performance

Data Output Messages Notifications

| | category text | total_sales numeric | total_profit numeric |
|---|---|---|---|
| 1 | Technology | 836154.03 | 145454.95 |
| 2 | Furniture | 741999.80 | 18451.27 |
| 3 | Office Suppli... | 719047.03 | 122490.80 |

Sales performance varies across product categories.

**Technology generates the highest revenue ($836K) and also delivers the strongest profitability ($145K), making it the most valuable product category for the business.**

Although **Furniture produces high revenue ($742K), it generates significantly lower profit ($18K), suggesting potential issues such as high discounting, logistics cost, or pricing inefficiencies.**

**Office Supplies shows balanced performance**, contributing strong revenue ($719K) with relatively healthy profit ($122K).

These findings suggest that management should:

- prioritize Technology category expansion
- review cost structure and pricing strategy in Furniture
- maintain steady growth in Office Supplies

## Discount Impact on Profitability

Data Output Messages Notifications

| | discount_level text | total_sales numeric | total_profit numeric |
|---|---|---|---|
| 1 | No Discount | 1087908.47 | 320987.60 |
| 2 | Low Discount | 846522.24 | 100785.47 |
| 3 | Medium Discou... | 298541.41 | -58817.00 |
| 4 | High Discount | 64228.74 | -76559.05 |

Discounting strategy has a significant impact on business profitability.

Orders with **no discount generate the highest total profit of $320.9K from $1.09M in revenue**, indicating that full-price sales are the most profitable transactions.

While **low discounts still generate positive profit ($100.8K from $846K sales)**, profitability begins to decline compared to full-price sales.

However, **medium discounts lead to overall losses (-$58.8K)** despite generating $298K in revenue, suggesting that aggressive discounting reduces margins below sustainable levels.

The situation worsens for **high discounts, which produce the largest losses (-$76.6K) from only $64K in sales**, indicating that heavy discounting is both unprofitable and ineffective in driving meaningful revenue growth.

These findings suggest that:

- excessive discounting erodes profit margins
- medium and high discount levels should be carefully reviewed
- pricing optimization strategies may significantly improve overall profitability

## Top Selling Products Performance

The analysis of top-selling products shows that revenue performance varies significantly across individual products, and high sales volume does not always translate into strong profitability.

The **Canon imageCLASS 2200 Advanced Copier** emerges as the best-performing product, generating **$61.6K in sales and $25.2K in profit**, making it both the highest revenue contributor and the most profitable item in the dataset.

Other strong performers include the **Fellowes PB500 Electric Punch Binding Machine**, which generated **$27.5K in sales and $7.8K in profit**, as well as the **Hewlett Packard LaserJet 3310 Copier**, contributing **$18.8K in sales and $7.0K in profit**.

However, several high-revenue products demonstrate weak or negative profitability. For example, the **Cisco TelePresence Videoconferencing Unit** generated **$22.6K in sales but resulted in a loss of $1.8K**, while the **GBC DocuBind P400 Electric Binding System** also recorded negative profit despite significant revenue contribution.

These findings suggest that while certain products drive strong revenue growth, pricing strategy, discounting, or cost structure may reduce their profitability. The business should therefore prioritize promoting high-margin products while reviewing pricing and cost efficiency for high-revenue but low-profit items.

## Monthly Sales Trend Analysis

Monthly performance analysis reveals strong seasonality in business revenue and profitability. Sales levels are generally lower at the start of each year and begin to increase mid-year before reaching peak levels in the final quarter.

The highest monthly revenue is recorded in November 2018 (~$118K), with similar spikes observed in Q4 periods across multiple years. This pattern indicates that demand is significantly higher toward year-end.

Additionally, certain months show negative profitability despite moderate sales volumes, suggesting the impact of discounting strategies or high operational costs.

Overall, the results indicate that the business should prioritize inventory planning, marketing investment, and staffing strategies toward high-demand periods, particularly in Q4.

## Negative Profit Products Analysis

Negative profit analysis shows that several products generate substantial revenue while still destroying profitability.

The **Cubify CubeX 3D Printer Double Head Print** is the worst-performing product, generating **$11.1K in sales but a loss of -$8.9K in profit**. Other major loss-making products include the **Lexmark MX611dhe Monochrome Laser Printer** with **$16.8K in sales and -$4.6K in profit**, and the **Cubify CubeX 3D Printer Triple Head Print**, which produced **$8.0K in sales but a loss of -$3.8K**.

This pattern suggests that certain products may be suffering from excessive discounting, poor pricing strategy, high procurement costs, or weak margin control. In some cases, products with strong revenue contribution still reduce total business profitability.

These findings indicate that management should review loss-making products individually, reassess pricing and discount structures, and consider whether some products should be repositioned, renegotiated with suppliers, or removed from active promotion.

## Sub-Category Profitability Analysis

Data Output Messages Notifications

| | sub_category text | total_sales numeric | total_profit numeric |
|---|---|---|---|
| 1 | Tables | 206965.53 | -17725.48 |
| 2 | Bookcases | 114880.00 | -3472.56 |
| 3 | Supplies | 46673.54 | -1189.10 |
| 4 | Fasteners | 3024.28 | 949.52 |
| 5 | Machines | 189238.63 | 3384.76 |
| 6 | Labels | 12486.31 | 5546.25 |
| 7 | Art | 27118.79 | 6527.79 |
| 8 | Envelopes | 16476.40 | 6964.18 |
| 9 | Furnishings | 91705.16 | 13059.14 |
| 10 | Appliances | 107532.16 | 18138.01 |
| 11 | Storage | 223843.61 | 21278.83 |
| 12 | Chairs | 328449.10 | 26590.17 |
| 13 | Binders | 203412.73 | 30221.76 |
| 14 | Paper | 78479.21 | 34053.57 |
| 15 | Accessories | 167380.32 | 41936.64 |
| 16 | Phones | 330007.05 | 44515.73 |
| 17 | Copiers | 149528.03 | 55617.82 |

Sub-category profitability analysis reveals that not all high-revenue sub-categories contribute positively to business performance.

**Tables** is the weakest-performing sub-category, generating **$207.0K in sales but a loss of -$17.7K in profit**, making it the largest profitability concern in the business. **Bookcases** also underperform, producing **$114.9K in sales but a loss of -$3.5K**, while **Supplies** records a smaller but still negative profit of **-$1.2K**.

On the other hand, several sub-categories show strong profitability. **Copiers** delivers the highest profit at **$55.6K**, followed by **Phones ($44.5K)** and **Accessories ($41.9K)**. These categories combine strong revenue with healthy profit contribution.

These findings suggest that the business should closely review the pricing, discounting, and cost structure of underperforming sub-categories such as Tables and Bookcases, while continuing to prioritize high-margin categories like Copiers, Phones, and Accessories.

# BUSINESS RECOMMENDATIONS

The analysis suggests several strategic actions to improve overall business performance and profitability.

First, management should **prioritize investment in high-performing regions such as the West and East**, while conducting deeper cost and pricing reviews in the Central region where profitability is relatively weak despite moderate sales volume. Targeted marketing and expansion initiatives may also help stimulate demand growth in the South region.

Second, the company should **optimize discounting policies**, as medium and high discount levels are shown to significantly reduce profitability. Implementing stricter discount approval processes or data-driven pricing strategies could help protect margins while maintaining sales volume.

Third, the business should **focus on expanding high-margin product categories such as Technology**, while reviewing operational costs and pricing strategy in the Furniture category, where revenue is high but profit contribution remains low.

Fourth, **loss-making products and sub-categories such as Tables and certain printer models should be carefully evaluated**. Management may consider

renegotiating supplier costs, adjusting pricing, reducing promotional focus, or discontinuing persistently unprofitable products.

Finally, the strong **seasonal sales spikes in Q4 suggest the need for proactive inventory planning, staffing adjustments, and marketing campaigns ahead of peak demand periods**. Aligning operational strategy with seasonal demand patterns can significantly improve both revenue and profitability outcomes.

# TECHNICAL SKILLS DEMONSTRATED

This project demonstrates practical business data analytics skills across data preparation, database querying, analytical modeling, and dashboard development.

**SQL & Database Skills**

- Data cleaning and structured data import into PostgreSQL
- Aggregation analysis using GROUP BY, ORDER BY, CASE statements
- Profitability segmentation and time-series analysis
- Creation of reusable SQL analytical views
- Query optimization for dashboard integration

**Business Intelligence Skills**

- Professional dashboard design in Power BI
- KPI development and executive-level layout structuring
- Interactive filtering and dynamic report exploration
- Time trend visualization and categorical performance comparison
- Profitability risk identification through visual analytics

**Analytical Thinking**

- Identification of revenue drivers and profitability risks
- Seasonality analysis for demand forecasting
- Discount impact evaluation on business margins
- Product-level performance diagnosis
- Strategic recommendation development based on data insights

# PROJECT FILES

- Cleaned dataset (CSV)
- PostgreSQL database scripts
- SQL analytical query files
- SQL view creation scripts
- Power BI dashboard file (.pbix)
- Project documentation (Notion / README)
- Dashboard screenshots
- GitHub repository link